



Progressive damage analysis for multiscale modelling of composite pressure vessels based on Puck failure criterion

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ABSTRACT

This study proposed a method for progressive failure analysis to investigate the failure modes of 35 MPa Type III composite pressure vessels during hydraulic burst tests. Representative volume elements (RVEs) were introduced, and the degraded elastic parameters of composite, which include fibre and matrix failure modes, were calculated by the finite element analysis (FEA) method to obtain the macroscopic stiffness degradation parameters. The Puck failure criterion was adopted, and the effective elastic parameters calculated from RVEs was introduced by the UDSLFD subroutine to simulate the multiscale progressive damage of composite layers during hydraulic burst test. Combined with acoustic emission detection, composite microscopic photos and hydraulic burst test, the burst pressure and failure modes of the vessels were analysed. The results showed that meso-macroscopic finite element models could effectively predict the progressive damage of composite pressure vessel, such as matrix crack, interlaminar failure and fiber fracture under internal pressure loading. Finally, the predicted burst pressure was about 5.4% average difference of the actual experimental results. Besides, cylinder section of the vessels was the predicted blasting position, which was close to the actual tests. Thus, the proposed method can provide theoretical reference for the design and optimisation of composite pressure vessels.

1. Introduction

Filament-wound composite pressure vessels have been widely used in the fields of aerospace, aviation and vehicle-mounted hydrogen storage tanks due to their high structural efficiency. The composite winding layers of vessels provide mechanical strength to structures [1,2]. In recent investigations, researches were focus on predicting the burst pressure of vessels in order to further reduce the weight of the composite layers. A simple stress analysis under internal pressure could not easily obtained accurate predictions of the burst pressure of composite pressure vessels. Because it did not take into account the matrix crack, interlaminar failure and final fiber fracture of the composites during loading case. Therefore, the multiscale progressive damage analysis of the composite layers was proposed to determine precisely the burst pressure of composite pressure vessels [3–5]. Besides, several criteria were used to predict the complex damage mechanisms of composite materials such as maximum stress or strain, Hashin, Hoffmann, Tsai–Wu and Tsai–Hill and so on [6,7]. The World

Wide Failure Exercise (WWFE) compared failure criteria for composite materials and Puck failure criterion has high accuracy in predicting composite failure [8]. Puck failure criterion was established on the basis of Hashin failure criterion. The inter fiber failure (IFF) of Puck criterion contained three failure modes including the transverse tension (mode A), moderate transverse compression (mode B), and large transverse compression (mode C). Also, it was necessary to determine the angle of the fracture plane [9]. In particular, the Puck criterion was rarely employed to predict the damage of composite pressure vessels because of its extreme complexity and there were rarely appropriate commercial finite element analysis (FEA) codes that contained the Puck criterion as a default option.

Different stiffness degradation criteria exert considerable impact on the progressive damage of composite materials [10,11]. However, the stiffness degradation parameters based on macro laminates did not involve the mesoscale of the fibre/matrix and researches on the subject had been mostly restricted to continuous damage mechanics and composite material homogenisation theory [12]. Therefore, the real

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characteristics of composite damage could not be accurately explained. Furthermore, the parameters of stiffness degradation parameters could not easily be obtained, which consume considerable amounts of energy and financial resources. On the basis of the micromechanics of composite materials, the effect of fibre/matrix failure on the mechanical properties of composite materials could be studied by establishing a representative volume elements (RVEs) [13]. A number of methods for extracting data from meso-mechanical properties to reflect the properties of macro materials have been proposed by numerous researchers and the multiscale failure analysis method for composite materials based on finite element analysis (FEA) has been employed [14,15]. Blassiau et al. [16], Bunsell and Thionnet [17] and Camara et al. [18] performed global/local multiscale failure analyses of composite pressure vessels by carrying out failure analyses on RVEs. However, the progressive damage characteristics of the matrix were not fully considered in their models, and the interfacial properties of the fiber and matrix were not studied. P.F. Liu [19] and Ren M F [20] established microscopic stiffness degradation models through RVEs units to predict the burst pressure of composite pressure vessels. However, the vessels in their work were laminated structures, and large angle changes were noted between different filament winding layers. Therefore, the stiffness degradation model of a microstructure could characterise the fracture failure of the fiber and matrix, but it could not easily simulate the interlaminar failure between filament winding layers. In this case, the research on the microscopic-macroscopic multiscale progressive failure analysis of composite vessels including matrix -fibre damage and interlaminar failure was significant.

In the present study, A stiffness degradation model of the RVEs was instructed by different failure modes of the fibre and matrix separately and macroscopic stiffness degradation parameters of the composites could be obtained by using the FEA method. An accurate model of layer-by-layer filament winding composite was established and cohesive elements were introduced between filament winding layers to simulate the interlaminar failure of composite pressure vessels. On the basis of the Puck criterion, the UDSLFD subroutine was compiled to introduce the stiffness degradation parameters. In addition, the progressive damage analysis of the composite layers under internal pressure was carried out with the aid of Abaqus software to predict the burst pressure and blasting modes of the vessels. Finally, the composite damage mechanism results of the FEA were verified by the hydrostatic burst tests of the vessels and the damage signals, which were monitored by acoustic emission during the hydrostatic tests.

2. Micromechanical damage model and FEA based on RVEs

2.1. FEA model of RVEs

TORAY T700 CF/epoxy composite were introduced and material properties were listed in Table 1. A hexagon RVEs was established as shown in Fig. 1. RVEs consisted of two components: the matrix and fibre. The matrix was an isotropic material and the fibre was an orthotropic material. The fibre volume fraction of the RVEs was 60%. The length of the hexahedral volume element b was 0.173 mm, the width a was 0.1 mm, and the thickness h was 0.005 mm. The EasyPBC plug-in was used to apply reasonably symmetric boundary conditions and periodic boundary conditions to the RVEs model [21], and homogenisation theory was employed to perform the FEA and calculate the elastic parameters of the composite material.

2.2. FEA results of RVEs

A stiffness degradation model of the RVEs was instructed different failure modes of the fibre and matrix separately, such as fibre failure,

Table 1
Material properties of T700 CF and epoxy resin.

Mechanical parameters of carbon fibre	Value	Mechanical parameters of epoxy resin	Value
E_{f11}	230 GPa	E_m	3.2 GPa
E_{f22}	28 GPa		
E_{f33}	28 GPa		
μ_{f12}	0.23	μ_m	0.35
μ_{f23}	0.3		
G_{f12}	50 GPa	G_m	1.17 GPa
G_{f13}	50 GPa		
G_{f23}	10 GPa		

matrix failure and fibre-matrix failure. Different failure modes were corresponded to different stiffness reduction coefficients, as shown in Table 2. Then, the macroscopic elastic parameters of the composite materials under different failure modes was abtained by EasyPBC plug-in based on the FEA of the RVEs, as listed in Table 3. These parameters provided effective calculation data for the progressive damage analysis of composite pressure vessels.

3. FEA model of composite pressure vessel

Aluminium alloy (6061-T6) liner was used for the 35 MPa Type III composite pressure vessel. Its basic dimensions were shown in Fig. 2. The volume of the composite pressure vessel was 57 L, the working pressure was 35 MPa, and the design burst pressure must be greater than 70 MPa. The stress-strain curve of the aluminium alloy material was shown in Fig. 3, which was introduced into the FEA model.

The FEA model of the composite pressure vessel was established by Abaqus 6.14. In order to save computing resources, a 1/36 periodic symmetric model of the vessel was established to FEA calculation. Besides, layer-by-layer modelling was performed with different winding angles and variable layer thicknesses (Fig. 4). The winding layers of the composite pressure vessel were shown in Table 4. Cohesive elements with 0 thickness were used to simulate the interlaminar failure of the composite pressure vessel during the internal pressure process (Fig. 4). The basic calculation parameters were shown in Table 5 [22]. The element types of the aluminium alloy liner, composite layers and cohesive were C3D8R, C3D8H and COH3D8, respectively.

The winding angle along the cylindrical section remained constant. The helical winding angle over the dome was described as Eq. (1) [23].

$$\theta(r) = \sin^{-1} \left(\frac{r_0}{r} \right) \quad (1)$$

where r_0 was the radius at helical turnaround point; r was the radius at a given point on dome.

The thickness specified in the winding layout table was the thickness at the cylinder tangent line. For the helical layers on the dome part, the layer thickness gradually increased and then sharply rose near the polar openings [23].

$$t(r) = \frac{t_d \cos(\theta_d)}{\cos(\theta_r)} \cdot \frac{r_d}{r + 2 \cdot BW \cdot \left(\frac{r_d - r}{r_d - r_0} \right)^4} \quad (2)$$

where t_d was the thickness at the tangent line; θ_d was the winding angle at the tangent line; θ_r stands for the winding angle at radius; r_d represented the radius at the dome/cylinder tangent line; BW was the helical band width.

The Puck criterion defined by Eq. (3)–(10) was employed to analyse and predict the failure of the composite layers [24,25]. Four variables, STATEV 1–4, were defined to record four basic failure modes, such as fibre tensile failure, fibre compression failure, matrix tensile failure and matrix compression failure.

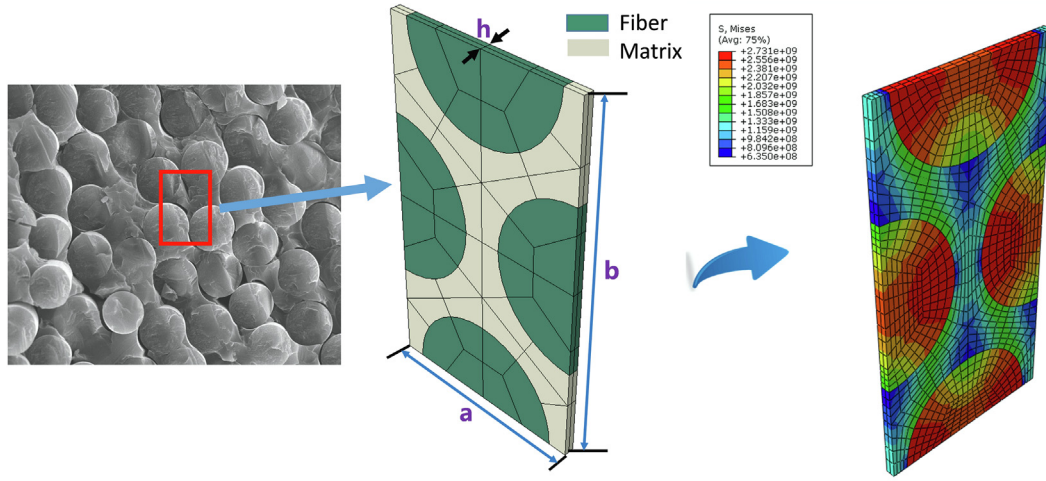


Fig. 1. FEA model of RVEs.

Table 2
Micro-stiffness reduction coefficient of RVE.

Loading-type failure category	Mesoscopic failure mode	Micro-stiffness reduction coefficient
Tensile failure	Fibre tensile failure	$E_{f11d} = 0.07E_{f11}$
		$E_{f22d} = E_{f33d} = 0.07 E_{f22}$
		$\mu_{f12d} = 0.07\mu_{f12}$
		$\mu_{f23d} = 0.07\mu_{f23}$
		$G_{f12d} = G_{f13d} = 0.07G_{f12}$
Compression failure	Matrix tensile failure	$G_{f23d} = 0.07 G_{f23}$
		$E_{md} = 0.2E_m$ $\mu_{md} = 0.2 \mu_m$
		$G_{md} = 0.2G_m$
		Ditto
		Ditto
Compression failure	Fibre compression failure	$E_{f11d} = 0.14E_{f11}$
		$E_{f22d} = E_{f33d} = 0.14 E_{f22}$
		$\mu_{f12d} = 0.14\mu_{f12}$
		$\mu_{f23d} = 0.14\mu_{f23}$
		$G_{f12d} = G_{f13d} = 0.14G_{f12}$
	Matrix compression failure	$G_{f23d} = 0.14 G_{f23}$
		$E_{md} = 0.4E_m$ $\mu_{md} = 0.4 \mu_m$
		$G_{md} = 0.4 G_m$
		Ditto
		Ditto
	Fibre-matrix compression failure	Ditto
		Ditto
		Ditto
		Ditto
		Ditto

The Puck fibre failure criterion was written as follows:

$$f_{E,FF} = \frac{1}{\pm R_{\parallel}^{t,c}} \left[\sigma_{11} - \left(V_{\perp\parallel} - V_{\perp\parallel f} \cdot m_{\sigma f} \frac{E_{\parallel}}{E_{\parallel f}} \right) (\sigma_{22} + \sigma_{33}) \right] \quad (3)$$

$$\text{with} \begin{cases} +R_{\parallel}^t & \text{for } [\dots] \geq 0 \\ -R_{\parallel}^c & \text{for } [\dots] \leq 0 \end{cases} \quad (4)$$

where $f_{E,FF}$ was the fiber failure stress exposure of the lamina; $\pm R_{\parallel}^{t,c}$ was the tensile and compressive strengths of the materials; σ_{11} , σ_{22} and σ_{33}

were the normal stresses in a unidirectional layer; $V_{\perp\parallel}$ and $V_{\perp\parallel f}$ were the major Poisson's ratios of the lamina and fibres, respectively; and $m_{\sigma f}$ was the stress magnification factor for transverse stresses in the fibres and was set as 1.1; $E_{\parallel}$ and $E_{\parallel f}$ were the moduli of the composite material and fibre direction, respectively, as shown in Table 6.

The Puck IFF criterion was written as follows:

For $\sigma_n \geq 0$

$$f_{E,IFF}(\theta) = \sqrt{\left[\left(\frac{1}{R_{\perp}^A} - \frac{p_{\perp\varphi}^t}{R_{\perp\varphi}^A} \right) \sigma_n(\theta) \right]^2 + \left(\frac{\tau_{nt}(\theta)}{R_{\perp\perp}^A} \right)^2 + \left(\frac{\tau_{n1}(\theta)}{R_{\perp\parallel}^A} \right)^2} + \frac{p_{\perp\varphi}^t}{R_{\perp\varphi}^A} \sigma_n(\theta) \quad (5)$$

For $\sigma_n < 0$

$$f_{E,IFF}(\theta) = \sqrt{\left[\left(\frac{p_{\perp\varphi}^c}{R_{\perp\varphi}^A} \right) \sigma_n(\theta) \right]^2 + \left(\frac{\tau_{nt}(\theta)}{R_{\perp\perp}^A} \right)^2 + \left(\frac{\tau_{n1}(\theta)}{R_{\perp\parallel}^A} \right)^2} + \frac{p_{\perp\varphi}^c}{R_{\perp\varphi}^A} \sigma_n(\theta) \quad (6)$$

where

$$\frac{p_{\perp\varphi}^{t,c}}{R_{\perp\varphi}^A} = \frac{p_{\perp\perp}^{t,c}}{R_{\perp\perp}^A} \cos^2 \varphi + \frac{p_{\perp\parallel}^{t,c}}{R_{\perp\parallel}^A} \sin^2 \varphi \quad (7)$$

$$\cos^2 \varphi = 1 - \sin^2 \varphi = \frac{\tau_{nt}^2}{\tau_{nt}^2 + \tau_{n1}^2} \quad (8)$$

$$R_{\perp\perp}^A = \frac{R_{\perp}^c}{2(1 + p_{\perp\perp}^c)} \quad (9)$$

where $R_{\perp}^{A,T}$ and $R_{\perp}^A$ were the tensile strength perpendicular to the fibre direction and the shear strength, respectively. $R_{\perp}^c$ was the compression strength perpendicular to the fibre direction. $R_{\perp\perp}^A$ was the fracture resistance of the action plane against its fracture due to transverse/transverse shear stressing. $p_{\perp\perp}^{t,c}$, $p_{\perp\parallel}^{t,c}$ were the inclination parameters.

Table 3
Macro elastic parameters in stiffness reduction.

Parameter	Undamaged	Fiber tensile failure	Fiber compression failure	Matrix tensile failure	Matrix compression failure
E_{f11}	139.29 GPa	11.02 GPa	20.681 GPa	138.26 GPa	138.52 GPa
E_{f22}	9.38 GPa	2.537 GPa	3.871 GPa	2.081 GPa	3.929 GPa
E_{f33}	9.38 GPa	2.537 GPa	3.871 GPa	2.081 GPa	3.929 GPa
μ_{f12}	0.2721	0.1926	0.1745	0.1876	0.2045
μ_{f23}	0.4236	0.2158	0.2471	0.1678	0.2142
G_{f12}	4.385 GPa	2.186 GPa	2.953 GPa	1.179 GPa	2.171 GPa
G_{f13}	4.385 GPa	2.186 GPa	2.953 GPa	1.179 GPa	2.171 GPa
G_{f23}	3.297 GPa	1.235 GPa	1.552 GPa	0.891 GPa	1.618 GPa

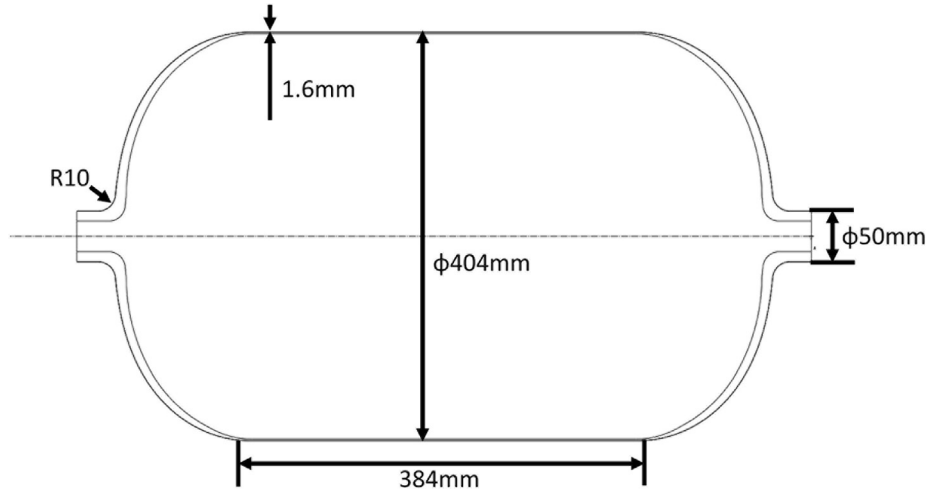


Fig. 2. Dimensions of aluminium alloy liner.

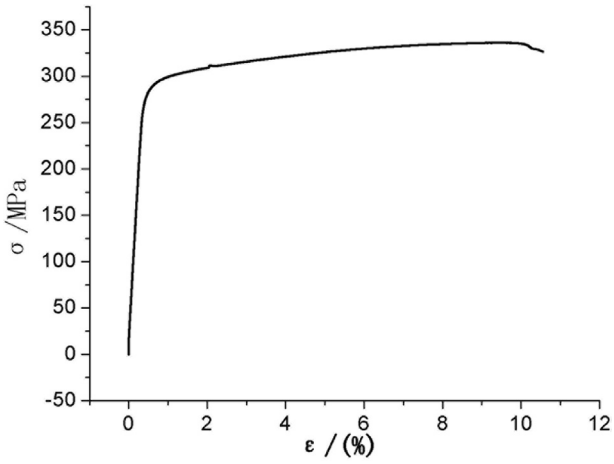


Fig. 3. Stress-strain curve of aluminium alloy.

$f_{EIFF}(\theta)$ was the failure effort or stress exposure of the inter-fibre failure of the lamina. When f was greater than 1, the fracture of the lamina occurs. If σ_n was the tensile stress, then it promoted IFF by assisting the shear stresses; if σ_n was the compressive stress, then it delayed IFF by raising the fracture resistances against shear fracture. Therefore, separate equations were used herein to evaluate IFF under tensile and compressive stresses σ_n . All the strength values were listed in Table 6.

The probability of failure was determined by the stress risk factor f_E on the action plane according to Puck criterion [26]. The plane with the largest stress risk factor was the failure plane. Thus, the failure criterion needed to find the position of the failure plane during compilation. The action plane orientated at the angle θ_{fp} was the fracture plane where the highest risk of fracture occurs. The angle was determined by calculating $f_{EIFF}(\theta)$ for all planes with angles ranging from 0° to 180° at 1° increments. Fracture can be easily predicted in the plane with the largest stress exposure.

$$[f_{EIFF}(\theta)]_{\max} = f_{EIFF}(\theta_{fp}) \quad (10)$$

The inclination parameters $p_{\perp\parallel}^{t,c}$, $p_{\perp\perp}^{t,c}$ were listed in Table 6.

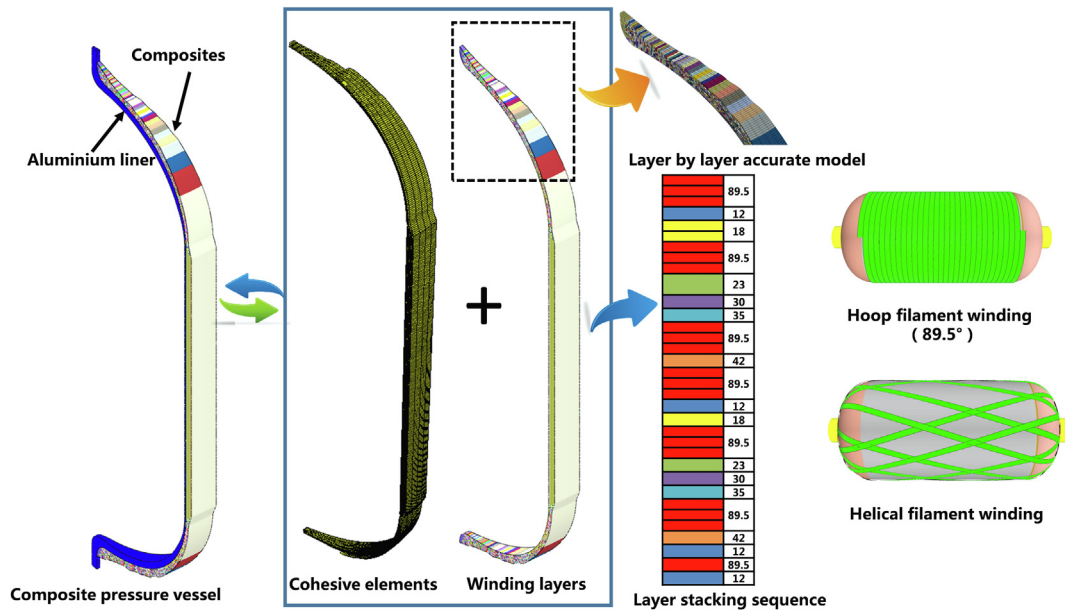


Fig. 4. FEA models of composite pressure vessels.

Table 4
Design Layers of composite pressure vessel.

Parameter	Number of layers corresponding to the winding angle							Total number of layers
	12°	18°	23°	30°	35°	42°	89.5°	
Winding angle	12°	18°	23°	30°	35°	42°	89.5°	
Number of layers	8	6	4	4	4	4	38	68

Table 5
Cohesive properties [22].

Properties	Values
Modulus E_0 (GPa)	3.0
Passions' ratio μ	0.35
Normal strength (N)	60
Shear strength $S_s = S_t$ (MPa)	85
Normal toughness G_{Ic} (N/mm)	0.28
Shear toughness G_{IIc} (N/mm)	0.79

Table 6
Parameters of Puck criterion.

NO	Parameter	Value
1	$P_{\perp}^I$	0.3
2	$P_{\perp}^C$	0.35
3	$P_{\perp}^{I,C}$	0.3
4	$E_{ f}$	230GPa
5	$E_{ }$	139.29GPa
6	$R_{ }^I$	2400 MPa
7	$R_{ }^C$	1400 MPa
8	$R_{\perp}^{AT}$	72 MPa
9	$R_{\perp}^A$	92 MPa
10	$R_{\perp}^C$	175 MPa

The Puck criterion was written in FORTRAN code as the user-defined UDSLFD subroutine for composite failure analysis. The maximum stress risk factor and fracture failure surface of all failure modes were determined by the Puck failure criterion. When the stress risk factor was greater than 1, failure occurred, and the material data after the failure were obtained. One or more failure modes can occur in one area, and the real stress state of the material after stress failure can be simulated by a damage variable diagram. The flow chart of the main program was shown in Fig. 5.

4. Experiments

4.1. Acoustic emission (AE) tests of composite pressure vessels

150 kHz sensors were used to detect composite pressure vessels. The detection threshold was set as 55 dB. The sensor layout was a combination of column positioning and area positioning. As shown in Fig. 6, five sensors were arranged at different parts of the vessels. No. 1 sensor was located at the upper nozzle of the composite pressure vessel, and No. 2–5 sensors were located at the upper and lower equators of the composite pressure vessel.

4.2. Hydraulic burst tests of composite pressure vessels

On the basis of the layer stacking sequence shown in Table 4, composite pressure vessels were fabricated with TORAY T700 CF/epoxy composite by wet winding technology. As a result of the equal polar openings of the vessels, the geodesic winding path was employed during the filament winding process. In order to reduce the thickness accumulation on the dome parts and improve the structural efficiency of the composite pressure vessels, the variable polar radii was utilised

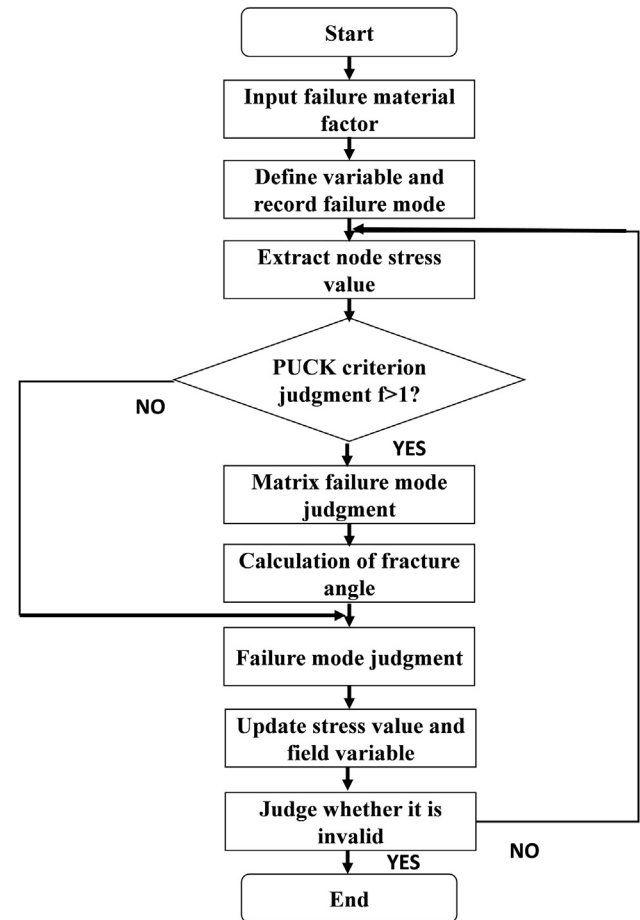


Fig. 5. UDSLFD subroutine flowchart based on Puck criterion.

in filament winding, as shown in Fig. 7. After the fabrication of the composite pressure vessels, the burst pressure was obtained by the hydraulic burst test.

5. Results and discussion

5.1. FEA results of composite pressure vessels

On the basis of the FEA models established by Abaqus, the progressive damage analysis of the composite pressure vessels was carried out to predict burst pressure by using the subroutine derived from Puck failure criterion, in which micromechanical material stiffness degradation parameters were introduced. As shown in Fig. 8(c), under the working pressure of 35 MPa, matrix tensile failure occurred near at the cylinder dome conjunction. As internal pressure increased to 60 MPa, the matrix tensile failure kept evolve gradually to other regions. However, with the internal pressure increasing to 76 MPa, the middle part of the cylindrical section showed a large deformation, the whole cylinder bulged outward and the whole section of the cylinder was appeared to serious matrix tensile failure. Fibre failure under

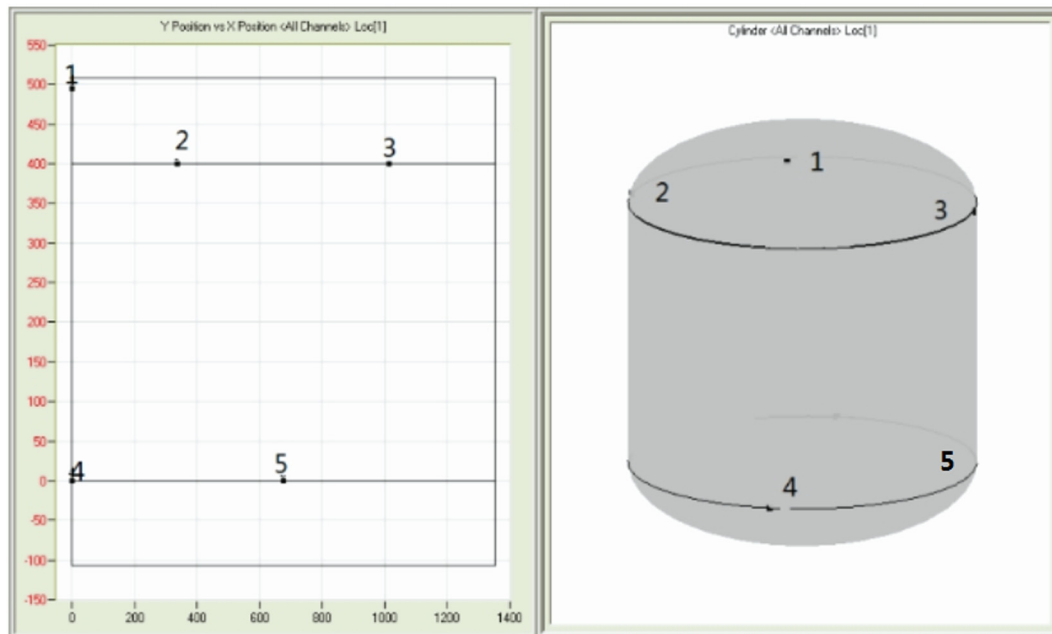


Fig. 6. Location of sensors in AE monitoring.

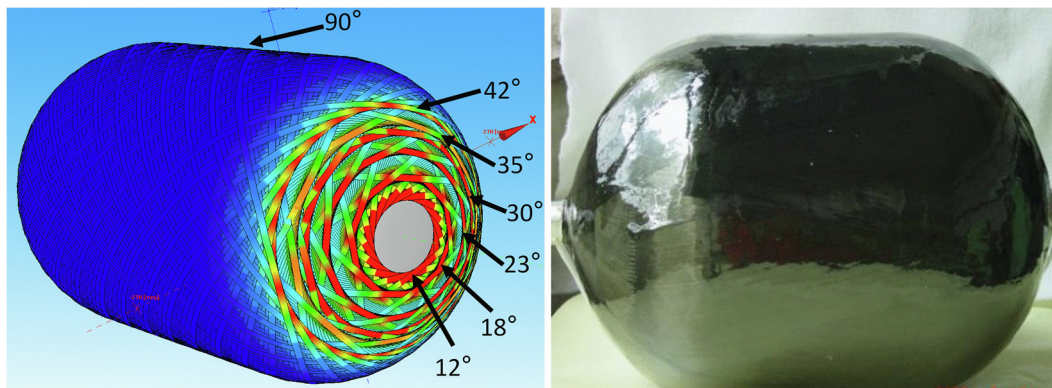


Fig. 7. Fabrication of composite pressure vessels.

tension was undetected under the internal pressure of 60 MPa as shown in Fig. 8(a). However, with the internal pressure increasing to 76 MPa, numerous fibre fracture exhibited in the bulge deformation area of the cylindrical section owing to the large tension generated in the composite layers (Fig. 8(a)). Besides, fibre fracture gradually penetrated from the outside to the inside. Nevertheless, fibre failure under compression was not found and small areas of matrix compressive failure was shown in Fig. 8(b) and (d). For the composite pressure vessel, the composite layers were accustomed to large loads; therefore, continue fibres played an important role in the presence of internal pressure [27]. When a large quantity of fibre fracture was happened in composite layers, composite pressure vessels would lose their ability to withstand internal pressure. So, matrix and fiber tensile failures were the main damage of composite pressure vessels at about 76 MPa internal pressure.

The cohesive elements were introduced into the layers between different winding angles to simulate the interlaminar failure of the composite pressure vessel, as depicted in Fig. 9. The cohesive elements showed apparent damage in the cylinder dome conjunction of the composite pressure vessel under 35 MPa internal pressure, which indicated interlaminar delamination failure between the filament winding

layers. As the internal pressure increased to 60 MPa, the delamination failure gradually expanded to the dome parts of the vessel. When internal pressure increased to 76 MPa, more delamination failure of the composites occurred in the cylindrical section than that in dome section. Combined with the FEA results of Fig. 8, with the increase of internal pressure, matrix failure of the composite pressure vessels generally occurred, followed by interlaminar delamination failure in the same area and finally expanding to fibre fracture; such process was consistent with the progressive damage process of composite materials [28]. Therefore, the meso-macroscopic model of the composite pressure vessel and the method for combining interlaminar cohesive elements in the present study could accurately simulate the matrix, fibre and interlaminar failure of the composite pressure vessel.

In order to predict the accurate burst pressure of the composite pressure vessel, the pressure-displacement curve of the composite pressure vessel in the FEA was drawn by selecting the element with the largest displacement of the cylindrical section, as shown in Fig. 10. The displacement of the cylinder section gradually increased with the internal pressure and the first linear curve denotes the initial linear-elastic behaviour of the materials. When the internal pressure increased to 60 MPa, the displacement curve began to show an obvious

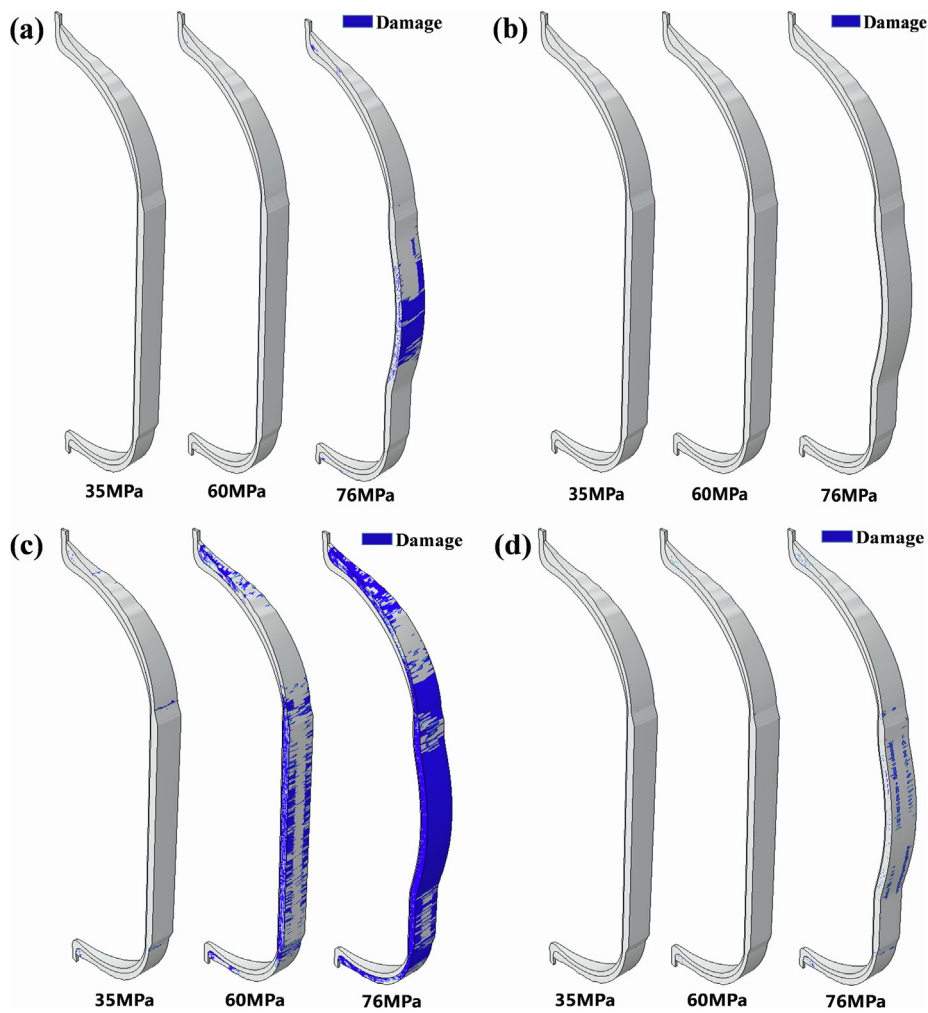


Fig. 8. Progressive damage of vessels:(a) Fibre tensile, (b) Fibre compression, (c) Matrix tensile, (d) Matrix compression.

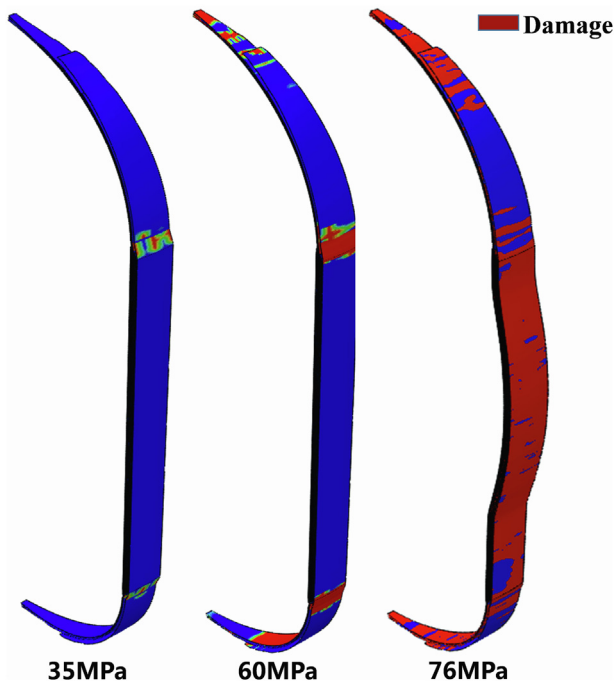


Fig. 9. Interlaminar failure of cohesive elements.

inflection point, indicating the decreases in the stiffness of the composite layer on the cylindrical section. It was mainly due to the progressive damage of the carbon fiber/epoxy composites. Nevertheless, as the internal pressure increased to about 76.6 MPa, the displacement increased sharply, thus indicating the fibre failure of the entire composite layers on the cylindrical section. Therefore, the stiffness of the composite layer decreased sharply, leading to the rapid increase of displacement. However, the internal pressure almost never changed. So, the predicted burst pressure of composite pressure vessels was about 76.6 MPa (Fig. 10). As fibre was subjected to the heaviest load in the composite layers, its failure in a large area would cause the composite layers to fail and reduce the load-bearing ability [29]. Additionally, the failure of the matrix and interlaminar could lead to a decrease of the entire stiffness of composites. However, such condition exerts little effect on the bearing capacity of the composite pressure vessel.

5.2. AE monitoring and microscopic analysis

The damage signals of the vessel were monitored by AE during the hydraulic burst test and the red points shown in Fig. 11 were represented the damage signals. According to previous studies, three kinds of predominant AE damage signals were observed in fibre-reinforced composites: matrix cracking, fibre-matrix debonding and fibre fracture [30]. As shown in Fig. 11, with the increase of internal pressure ranging from 35 MPa to 60 MPa, the damage signals gradually increased, thus indicating the progressive damage of the composite pressure vessel. Under the pressure of 44 MPa, the AE damage signals

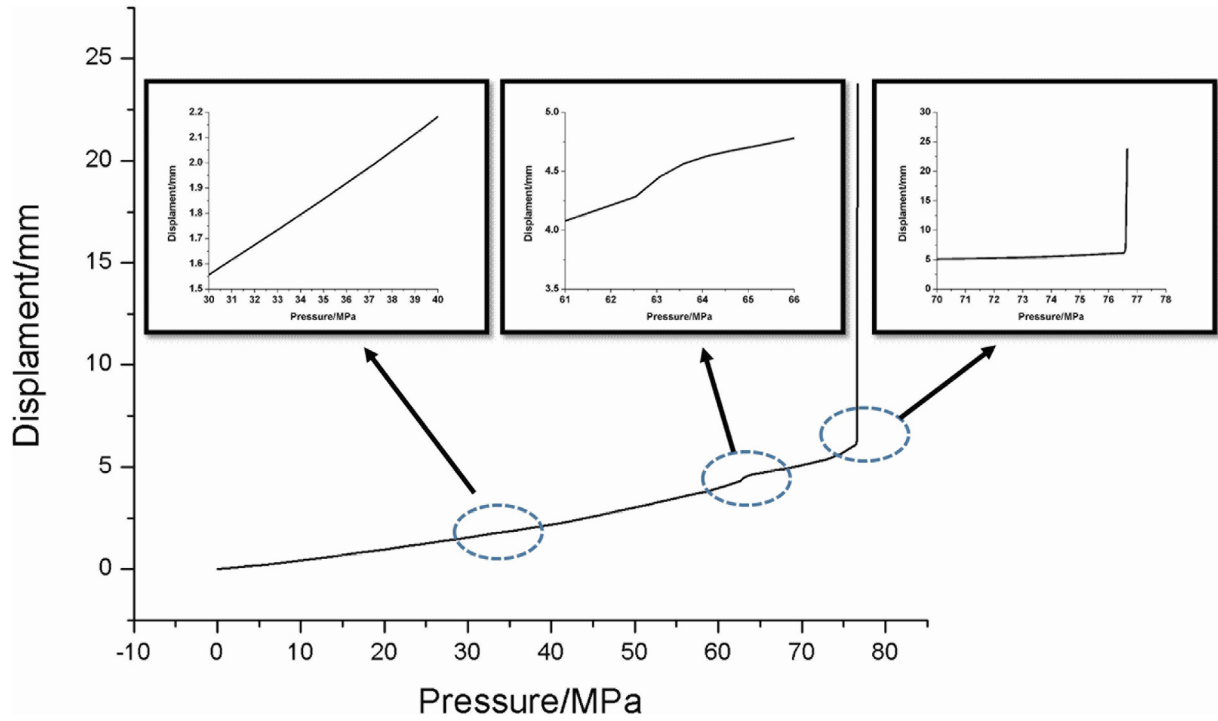


Fig. 10. Internal pressure-displacement curve of composite pressure vessel.

of the vessel were focused on the cylindrical and cylinder dome conjunction region. However, with the pressure increasing to 60 MPa, the AE damage signals became concentrated on the cylinder section. To prevent damage from AE probes, the probes were removed from the vessels after 60 MPa, so that the latter damage signals could not be detected. The monitoring results of the AE damage signals are basically consistent with the progressive damage results of the composite pressure vessel based on FEA. By analysing the increased position of AE damage signals, the cylindrical section of the composite pressure vessel would be the predicted blasting position.

To further study the damage mechanism of the composite pressure vessels during the hydraulic burst test, the surface cracks of the vessels were observed. The vessel under an internal pressure greater than 60 MPa were dissected and sampled for metallographic and scanning electron microscope observation. Fig. 12 showed the presence of numerous matrix cracks at the cylinder dome conjunction of the composite pressure vessel under a pressure of 44 MPa. Besides, the direction of the surface matrix crack tended to follow the length direction of the outermost fibres. Fibre fracture and the matrix crack of the cylindrical section were not obvious. However, as the internal pressure increased to 60 MPa, the entire vessel was covered with matrix crack.

But fiber fracture was also not obviously found (Fig. 13). It was indicated that the cylinder dome conjunction of the composite pressure vessel may be the first site of matrix crack, which were basically consistent with the results of the FEA (Fig. 8(c)).

To further investigate the damage mechanism of the composite pressure vessel, the vessel completed the 60 MPa hydrostatic test was dissected to further corresponding analysis. As shown in Fig. 14 (a), the composite matrix was migrated from the inner to the outer layers during the winding and curing processes of the composite pressure vessel [31], resulting in a matrix layer on the outer surface. Because of the great difference modulus between the fiber and matrix, the surface matrix layer was led to generating stress concentration during low internal pressure, which could result in matrix crack [32]. Fig. 14(b) indicates that matrix crack occurred in the internal filament winding layers on the cylindrical section at the internal pressure of 60 MPa. However, the propagation of matrix crack was stopped when it expanded to two different winding angle layers. Therefore, matrix crack did not spread along the interlaminar of the composites, which indicated little interlaminar failure occurred on the cylindrical section at 60 MPa. However, more matrix crack was found at the cylinder dome conjunction of the composite pressure vessel than that in the

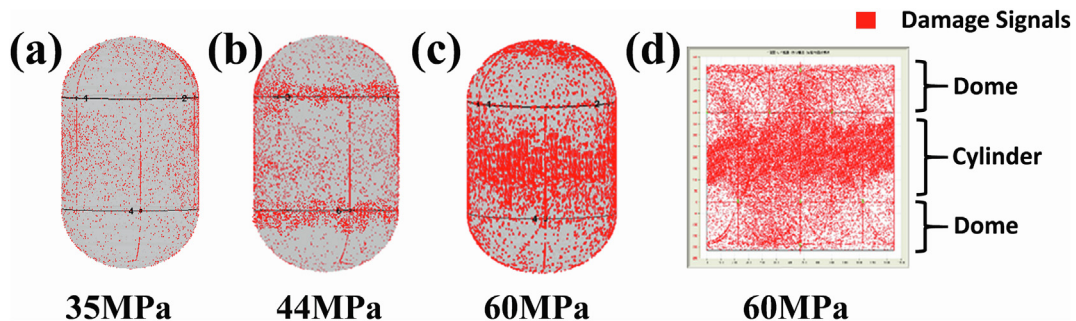


Fig. 11. AE monitoring during hydraulic burst tests.

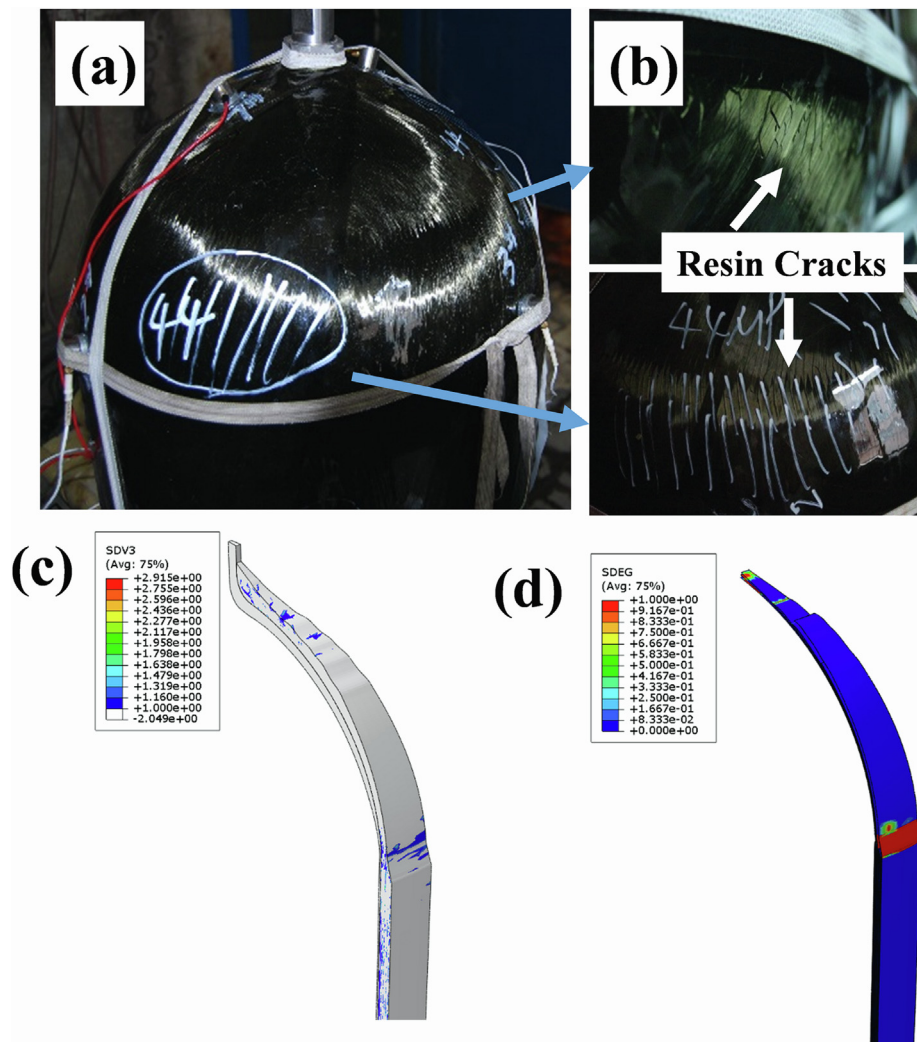


Fig. 12. Surface cracks of composite pressure vessel at 44 MPa.

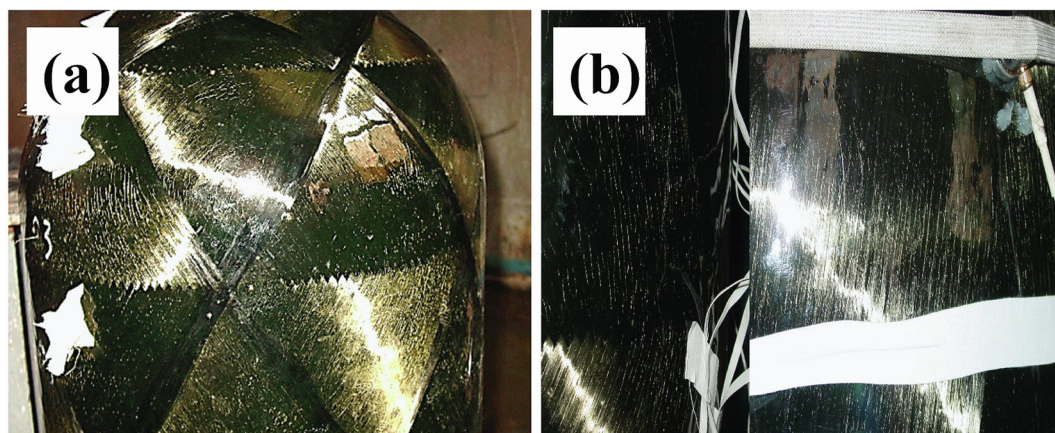


Fig. 13. Surface cracks of composite pressure vessel at 60 MPa.

cylindrical section (Fig. 15). Besides, interlaminar delamination failure was obviously found, as shown in Fig. 15. It was mainly because the layer thickness on the transition section of the composite pressure vessel was not enhanced by the loop layers. Besides, the cylinder dome conjunction of the vessel existed geometric discontinuities, which gave rise to bending moments. Out-of-plane shear forces should be devel-

oped to balance bending moment. So, the transition section was the weak link in the strength design of the composite pressure vessel. Besides, the porosity defects of the dome section significantly increased relative to those on the cylinder position due to fibre accumulation at the transition position (Fig. 15). Therefore, the cylinder dome conjunction of the composite pressure vessel was prone to stress

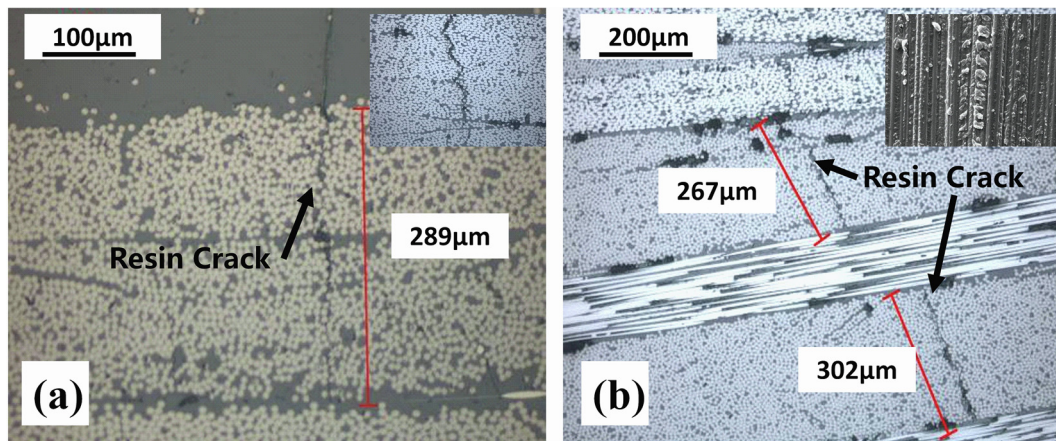


Fig. 14. SEM observations of the vessel's cylindrical section at 60 MPa.

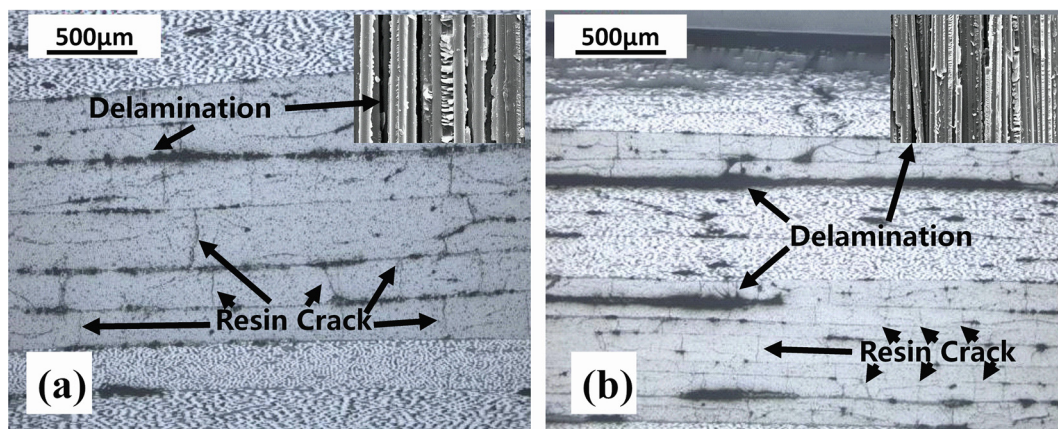


Fig. 15. SEM observations of the vessel's cylinder dome conjunction at 60 MPa.

concentration during the hydrostatic test leading to matrix crack and interlaminar delamination failure. AE monitoring and microscopic analysis of above results was consistent with the progressive damage results of finite element calculation, as shown in Figs. 8 and 9.

5.3. Hydraulic burst tests of composite pressure vessels

According to the layer stacking sequence listed in Table 4, five composite pressure vessels were fabricated to carry out hydraulic burst tests. The burst pressure of composite pressure vessels was 78.5–83.2 MPa due to process fluctuations in the manufacturing process, while the predicted result was 76.6 MPa, as listed in Table 7. So, it was about 5.4% average difference of the burst pressure between the actual tests and predicted result. It was indicated the burst pressure of the composite pressure vessel calculated by the FEA was in good agreement with the experimental results. So, the method of calculating

the elastic coefficient of stiffness degradation and the progressive damage model based on the Puck criterion were reliable.

The blasting modes of the composite cylinder were shown in Fig. 16. All the entire cylindrical section of the composite pressure vessels was mostly damaged, while the dome parts were comparatively integrated, which indicated that the five composite pressure vessels presented regular failure modes [33]. Besides, the blasting modes of the composite pressure vessel were in good agreement with the simulation results as shown in Fig. 16(d). Fig. 17 showed that the fibre fracture of the hook and helical filament winding layers of composite pressure vessel at 76.6 MPa. It could lead to the complete failure of the composite pressure vessel. The areas of fiber fracture in the hoop layers was much more than that in the helical layers at 76.6 MPa. It was indicated the fibre fracture of the hoop layers firstly occurred as the internal pressure increased. It gave rise to the decrease of the entire stiffness of the cylindrical section and the stress of the composite

Table 7
Burst pressure of composite pressure vessels.

NO	Experimental results/MPa	Predicted result/MPa	Percent difference/%	Percent difference on average/%
1#	82.1	76.6	7.2	5.4
2#	79.5		3.8	
3#	83.2		8.6	
4#	78.5		2.5	
5#	80.5		5.1	

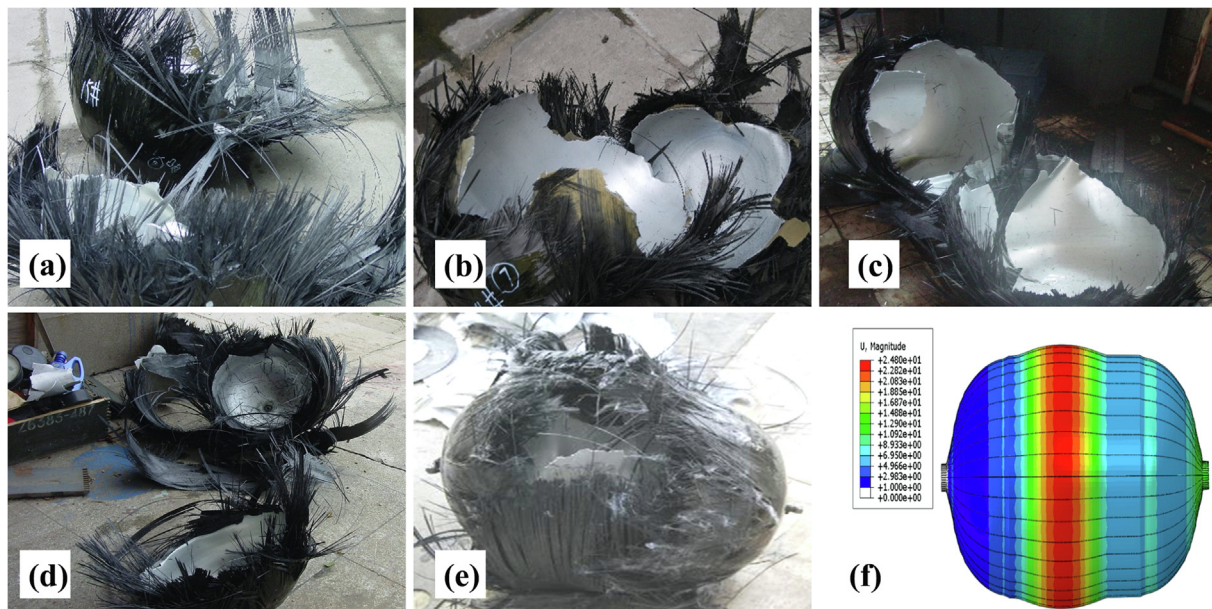


Fig. 16. Blasting modes of composite pressure vessels: (a) 1#, (b) 2#, (c) 3#, (d) 4#, (e) 5#, (f) FEA predicted result.

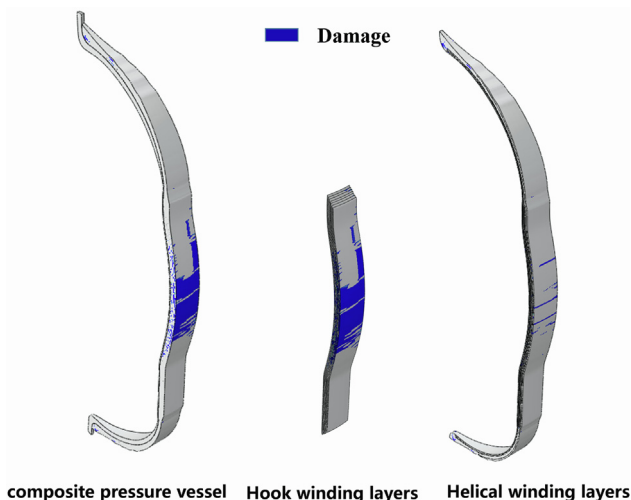


Fig. 17. Fibre fracture of composite pressure vessel at 76.6 MPa.

layers were redistributed. Finally, when the helical layers appeared fiber fracture, composite pressure vessels were prone to burst.

6. Conclusions

In present study, RVEs was established to introduce the failure modes of the fibre and matrix fracture. Then the macroscopic stiffness degradation parameters were calculated using the FEA method, which was compiled in UDSLFD subroutine. The progressive damage model based on the Puck criterion was established with the aid of Abaqus software to predict the burst pressure of the composite pressure vessels. Finally, cohesive elements were used to simulate the interlaminar damage of the vessels. To further validate the FEA results, composite pressure vessels were fabricated and conducted hydraulic burst tests. Furthermore, AE and microscopic observations were used to analyse the damage models of the composite layers.

The results indicated that the RVEs model established by micromechanics and FEA method could be employed to calculate the stiffness

degradation parameters of composite materials, which were used to determine the key parameters for the progressive damage analysis of composite pressure vessels. In this way, the meso-macroscopic relationship of composite materials could be established and the tests of the material properties was significantly reduced. The matrix crack and fibre fracture of composite pressure vessels could be investigated by the stiffness degradation model based on the Puck failure criterion. Cohesive elements introduced between filament winding layers could accurately simulate interlaminar failure. Based on the multiscale modelling of the composite pressure vessels, it was useful to study the damage mechanism of composite cylinder in the process of internal pressure, which was agree with the AE damage signals' monitoring and anatomic observation of composite materials. Finally, the burst pressure and blasting modes of the vessels was in good agreement with the FEA result. In particular, it was about 5.4% average difference of the burst pressure between the actual tests and predicted result. So, the proposed method can effectively predict the progressive damage of composite pressure vessels.

However, due to limited conditions, tracking the AE damages of the composite pressure vessels at the last stage after 60 MPa was impossible. Further research on AE detection will be carried out to further optimise the finite element model of composite pressure vessels.

7. Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study

8. Author Contribution Statement

Song Lin planned the project and provided financial support for this research activity. Liuqing Yang helped to perform the experiments and edit the manuscript. Hui Xu implemented the numerical simulation and participated in drafting the manuscript. Xiaolong Jia implemented the experiments, in particular, the preparation of composite pressure vessels. Xiaoping Yang validated the formulations and participated in drafting the manuscript. Lei Zu formulated overarching research goals and aims.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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